Programme	:	B.Tech.[CSE]	**Semester**	:	Fall Semester 2026-2027
Course	:	Computer Networks	**Code**	:	CSE3006
Faculty	:	Dr.S.Periyanayagi	**Slot/ Class No.**	:	B11+B12+B13, BL2026270100338
Tutorial No.	:	1	**Max. Marks**	:	50 Marks
Date of Submission: 04.08.2026					

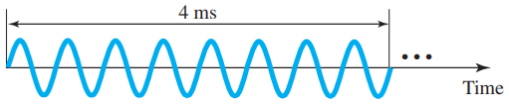
1. Assume a system uses five protocol layers. If the application program creates a message of 100 bytes and each layer (including the fifth and the first) adds a header of 10 bytes to the data unit, what is the efficiency (the ratio of application layer bytes to the number of bytes transmitted) of the system. [CO1, KL3] (5 Marks)

2. A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal and to sample the signal at twice the highest frequency (two samples per hertz). Assume that each sample requires 8 bits. What is the required bit rate? [CO2, KL3] (3 Marks)

3. An audio signal of frequency 5 kHz modulates a carrier signal of frequency 500 kHz. [CO2, KL3] (5 Marks)
   a. Calculate the upper sideband (USB) frequency.
   b. Calculate the lower sideband (LSB) frequency.
   c. Determine the bandwidth of the AM signal.

4. A digital communication system transmits data at a rate of 9600 bps. [CO2, KL3] (5 Marks)
   a. Calculate the baud rate for BPSK.
   b. Calculate the baud rate for QPSK.
   c. Calculate the baud rate for 8-PSK.
   d. Which modulation technique has the highest bandwidth efficiency?

5. What is the frequency of the signal? [CO1, KL2] (2 Marks)



6. A binary FSK system uses: [CO2, KL3] (5 Marks)
   Mark frequency = 120 kHz
   Space frequency = 100 kHz
   Bit duration = 0.5 ms
   For the bit sequence: 10110010
   a. Determine the frequency transmitted for each bit.
   b. Draw the corresponding FSK waveform.

7. Draw the graph of the NRZ-L scheme, NRZ-I, Manchester, Differential Manchester scheme using each of the following data streams. [CO1 & CO2, KL2] (10 Marks)

a. 00000000 b. 11111111 c. 01010101 d. 00110011

8. A sampled low-pass signal with a bandwidth of 200 KHz using 1024 levels of quantization.
   a. Calculate the bit rate of the digitized signal. [CO2, KL3] (5 Marks)
   b. Calculate the SNR_{dB} for this signal.
   c. Calculate the PCM bandwidth of this signal.

9. Calculate the baud rate for the given bit rate and type of modulation. [CO2, KL3] (5 Marks)
   a. 2000 bps, FSK
   b. 4000 bps, ASK
   c. 6000 bps, QPSK
   d. 36,000 bps, 64-QAM

10. If a periodic signal is decomposed into seven sine waves with frequencies of 1000, 2000, 4000, 5000, 7000, 8000 and 9000 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 20 V. [CO1, KL3] (5 Marks)

11. A periodic signal has a bandwidth of 40 KHz. The highest frequency is 80 KHz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude. [CO1, KL2] (5 Marks)

12. What are the propagation time and the transmission time for a 5-MB (megabyte) message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×108 m/s. [CO1, KL3] (5 Marks)

13. A communication system uses 64-QAM to transmit data at a rate of 48 Mbps.
   a. How many bits are transmitted per symbol? [CO2, KL3] (5 Marks)
   b. Calculate the required symbol (baud) rate.
   c. If the system is upgraded to 256-QAM, calculate the new baud rate for the same bit rate.
   d. Which modulation scheme is more bandwidth efficient?

14. What is the Nyquist sampling rate for each of the following signals? [CO2, KL3] (5 Marks)
   a. A low-pass signal with bandwidth of 200 KHz?
   b. A band-pass signal with bandwidth of 200 KHz if the lowest frequency is 100 KHz?

15. A digital communication system transmits binary data at a rate of 100 kbps using Manchester encoding. [CO2, KL3] (5 Marks)
   a. Determine the minimum signal rate (baud rate).
   b. Estimate the minimum required bandwidth.
   c. If the same data is transmitted using Polar NRZ-L, calculate the required bandwidth.
   d. Which encoding is more bandwidth efficient?

Course Outcome:
CO1: The students will be able to understand the basic principles of Communication, Networks and Satellite Transponders.
CO2: The students will be able to understand bit rate and baud rate.
KL1 – Remember, KL2-Understand, KL3-Apply